



MATHS · YEAR 11 · AQA

Non-Linear Simultaneous Equations

Simultaneous Equations · GCSE Higher (Grade 8–9)

NAME _____

DATE 06/06/2026

CLASS _____

Learning Objective

Solve a pair of equations where one is a straight line (linear) and one is a curve (quadratic or circle) using substitution. Most pairs have two answer points — find both x values and the y that goes with each. Write each answer as a coordinate (x, y).

Key Words

Simultaneous equations — equations that are true at the same time; the solution is where the graphs meet.

Substitution — replacing y (or x) using one equation so the other has a single unknown.

Non-linear — an equation whose graph is a curve (quadratic or circle).

Surd form — an exact answer left with a root sign, e.g. $(-1 + \sqrt{17})/2$.

Method — Five Steps

Step 1: Make y (or x) the subject of the straight-line equation.

Step 2: Substitute that into the non-linear equation.

Step 3: Move everything to one side so it equals 0.

Step 4: Solve the quadratic — factorise or use the formula.

Step 5: Put each value back to find its pair. Write answers as (x, y) and check.

Worked Example

Solve: $y = x^2 - 2x + 3$ and $y = x + 3$

Substitute: $x^2 - 2x + 3 = x + 3$

Rearrange: $x^2 - 3x = 0$

Factorise: $x(x - 3) = 0 \rightarrow x = 0$ or $x = 3$

Find y: $x = 0 \rightarrow y = 3$; $x = 3 \rightarrow y = 6$

Answers: (0, 3) and (3, 6)

Question 1

3 marks

Solve the simultaneous equations.

$y = x^2$ and $y = 2x + 3$

Question 2**3 marks**

Solve the simultaneous equations.

$$y = x^2 + 1 \quad \text{and} \quad y = x + 3$$

Question 3**3 marks**

Solve the simultaneous equations.

$$y = x^2 - 2x \quad \text{and} \quad y = x + 4$$

Question 4**3 marks**

Solve the simultaneous equations.

$$y = x^2 - 5x + 6 \quad \text{and} \quad y = x - 2$$

Question 5**4 marks**

Solve the simultaneous equations.

$$x^2 + y^2 = 25 \quad \text{and} \quad y = x - 1$$

Question 6**4 marks**

Solve the simultaneous equations.

$$x^2 + y^2 = 10 \quad \text{and} \quad x + y = 4$$

Question 7**5 marks**

Solve the simultaneous equations. Give your answers in surd form.

$$x^2 + y^2 = 9 \quad \text{and} \quad y = x + 1$$

Question 8**5 marks**

A rectangle has an area of 12 cm^2 and a perimeter of 14 cm .

By forming and solving a pair of simultaneous equations, find the length and width of the rectangle.

(Let the length be x and the width be y .)

Common Mistakes

- ✗ There are usually TWO answers, not one — find both x values (e.g. $x = 3$ and $x = -1$).
- ✗ Get everything on one side so it equals 0 first: $x^2 = 2x + 3$ becomes $x^2 - 2x - 3 = 0$.
- ✗ Watch the minus signs: $(x - 1)^2 = x^2 - 2x + 1$, not $x^2 - 1$.
- ✗ Each x has its own y — after finding x , substitute back to find the matching y .

How Did I Do?

How confident do you feel with this topic now? (tick one)

☐ Not yet ☐ Getting there ☐ Confident ☐ Really confident

- Which question did you find the hardest, and why?
- One method step I want to remember next time:
- One thing I will practise before the next lesson:

Teacher Key & Mark Scheme

Award method marks for a correct substitution and rearrangement to = 0, and accuracy marks for both coordinate pairs.

Q1 (3): (3, 9) and (−1, 1)

Q2 (3): (2, 5) and (−1, 2)

Q3 (3): (4, 8) and (−1, 3)

Q4 (3): (4, 2) and (2, 0)

Q5 (4): (4, 3) and (−3, −4)

Q6 (4): (1, 3) and (3, 1)

Q7 (5): $x = \frac{-1 + \sqrt{17}}{2}$, $y = \frac{1 + \sqrt{17}}{2}$ and $x = \frac{-1 - \sqrt{17}}{2}$, $y = \frac{1 - \sqrt{17}}{2}$

Q8 (5): length 4 cm, width 3 cm (from $xy = 12$ and $x + y = 7$)

Total: 30 marks.